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Design and Experimental Characterization of a Swirl-Stabilized Combustor for Low Calorific Value Gaseous Fuels

Low calorific value (LCV) gaseous fuels are generated as by-products in many commercial sectors, e.g., as mine gas or biogas. Their efficient exploitation can be a considerable source of primary energy. Typically, product gases from biomass are characterized by low lower heating values (LHVs) due to their high concentration of inert gases and steam. At the same time, their composition varies strongly based on the initial feedstock and may contain unwanted components in the form of tars and ammonia. These properties make the design of appropriate combustion systems very challenging and issues such as ignition, flame stability, emission control, and combustion efficiency must be accounted for. By employing a proprietary gas turbine burner at the TU Berlin, the combustion of an artificial LCV gas mixture at stoichiometric conditions has been successfully demonstrated for a broad range of steam content in the fuel. This work presents the stability maps and emissions measured with the swirl-stabilized burner at premixed conditions. It was shown that the flame location and shape primarily depend on the steam content of the LCV gas. The steam content in the fuel was increased until flame blow-out occurred at LHVs well below the target condition of 2.87 MJ/kg (2.7 MJ/m^3_N). The exhaust gas is analyzed in terms of the pollutants NO_x and CO for different fuel compositions, moisture contents, and thermal powers. Finally, OH^* measurements have been carried out in the flame. A simple reactor network simulation was used to confirm the feasibility of the experimental results. [DOI: 10.1115/1.4049642]

Introduction

Climate change is a driving force behind a restructuring of the energy sector toward a reduction of fossil fuel exploitation. The share of renewable energy has been increasing globally over the past years [1]. Many commercial processes also lead to the formation or accumulation of low calorific value (LCV) gaseous fuels, in the form of mine gas or product gas from steel manufacture. Moreover, waste products (e.g., biowaste, waste materials from forestry industry) can be processed to LCV gas. According to estimates, biogas can potentially provide up to 15% of Germany's primary energy demand [2]. In 2017, 8.5% of Germany's gross electricity consumption was covered from biogas including landfill gas and sewage gas [3].

In Germany, most of the biogas is exploited in combined heat and power plants employing reciprocating engines [4]. Typically, the latter offer efficiencies of around 40%, whereas the combined heat and power generation process achieves total efficiencies of around 90%. However, reciprocating engines require a minimum methane content of around 40%, so that LCV gases with a lower energy content cannot be utilized in many currently installed systems. The combination of gasification and combustion of biogas in (micro) gas turbines offers long-term potential for LCV gas exploitation [2].

Low calorific value gas that cannot be exploited in power or heat applications is often flared, thus contributing as a greenhouse gas and also providing no economic value [2]. The main difficulty in combusting low-lower heating value (LHV) gases (with LHVs of 3 to 20 times lower compared to methane [5]) in gas turbines is establishing a stable flame [2]. The gas turbine burner must offer an adequate design for that particular application. Using flameless combustion concept, the DLR FLOX combustor was successfully tested in a Turbec T100 microgas turbine with LCV gas with a

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Manuscript received September 1, 2020; final manuscript received September 21, 2020; published online December 9, 2021. Editor: Jerzy T. Sawicki.

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minimum LHV of 3.5 MJ/kg [6]. Similar LHV gas mixtures were also successfully burned in a modified Volvo VT4400 combustor, with combustion efficiencies above 99% and reasonable CO (< 20 ppm) and NO_x (< 9 ppm) emissions [7]. When ammonia was added to the product gas however, NO_x emissions were above 250 ppm. Based on their experimental results using a diffusion type flame and a product gas with a LHV of 4.46 MJ/m_N^3, the Alstom group concludes that even lower LCV gases are likely combustible using jet shear layer combustion [8]. When the same combustor was used at TU Delft, stable combustion with LHVs between 2.5 and 3.5 MJ/m_N^3 with combustion efficiencies >99% was reported [9]. NO_x formation has been analyzed using a Siemens V94.3A and LCV gases with LHVs down to 6.3 MJ/kg, achieving NO_x emissions of 67 ppm [2].

The high diluent content of noncombustible substances in LCV gas leads to a decrease of the chemical kinetic rates, which affects the flame physics in various ways [10]. The diluent content increases the specific heat of the fuel–air mixture, which decreases the adiabatic flame temperature [10], and as a result, the thermal efficiency of the combustion system. However, the low flame temperatures were found to decrease the NO_x emission levels, as the thermal NO_x path (typically the largest contributor to NO_x in modern gas turbines) is suppressed [8,9]. At the same time, the low hydrocarbon content was found to reduce the prompt NO_x formation [8]. This suggests that the combustion of LCV product gas may result in an overall low NO_x formation. Product gas from thermal gasification of biomass comprises more lighter components such as CO and H_2 compared to the content of CH_4. Further, it has been reported that the ratio of the combustible components has a significant effect on the NO_x emissions [2].

Nevertheless, due to a varying content of fuel-bound nitrogen in the LCV product gas, especially ammonia (NH_3), multiple groups report significant NO_x emissions, as a substantial fraction of NH_3 is converted to fuel NO_x. Many studies suggest that the presence of ammonia (NH_3) in the LCV product gas leads to increased NO_x emissions, with NO as the major pollutant species [9]. A full load simulation of the Alstom LVB combustor resulted in NO_x emissions <50 ppm for ammonia concentrations <400 ppm and conversion rates of 40–80% [8]. In an experimental study, conversion rates of approximately 30% were reported for ammonia concentrations between 2500 and 3800 ppm, with the remainder of the ammonia reacting to molecular nitrogen [9]. The same study concluded that the fuel NO mechanism due to NH_3 in the product gas is the dominant NO formation mechanism. The contributions of the thermal and prompt NO mechanisms were reported to be negligible, due to low adiabatic flame temperatures and the small concentration of hydrocarbon radicals. The study suggests further that the NH_3 to NO conversion rates decrease at elevated pressures, due to decreased concentrations of O and OH radicals at elevated pressures, however this was not confirmed experimentally yet. Similar results were obtained in an experimental study in an Alstom combustor, where NO was also observed as the major pollutant [11]. Conversion rates were expected to drop at elevated pressures. In a more recent experimental study in the so-called WINNOX TUD combustor, the same group reported much lower conversion rates between 40 and 5%, depending on the NH_3 concentration and the CH_4 content in the fuel gas [12]. They observed higher conversion rates for lower NH_3 concentrations and increased CH_4 content. They also reported a minimum conversion rate with a primary burner air number around 0.7, which translates to an equivalence ratio of 1.43. An increased conversion rate from NH_3 to NO_x for increased CH_4 content was also reported earlier by another group [13]. Additionally, the presence of CO_2 was found to increase CO formation in the exhaust gas [10].

The laminar burning velocity of the gas mixture depends on the methane and hydrogen content in the product gas [14]. The laminar burning velocity of the LCV product gas is expected to increase substantially if hydrogen is added to the fuel, since the laminar burning velocity of pure hydrogen–air mixtures is approximately 170 cm/s [15]. An experimental study of laminar burning velocities for two different generic biogas mixtures with normal air yielded 28 cm/s for a 66% CH_4/34% CO_2 mixture and 46 cm/s for a 33% CH_4/17% CO_2/40% C_3H_8/10% H_2 mixture (in percent volume) [16]. A dilution with CO_2 led to a decrease of the laminar flame speed as reported in Refs. [17] and [18]. The laminar flame speed was observed to decrease as well when the fuel was diluted with nitrogen, which also led to a shift of the peak flame temperature and velocity closer to stoichiometry [19].

The high diluent content was observed to lead to a narrower flammability limit compared to natural gas. Adding CO_2 was observed to increase the flammability limits due to an increased radiation absorption, while adding steam had a negative effect on the flammability [17].

The above-mentioned detrimental effects on the flame physics lead to technical challenges for the combustion of LCV gas in a gas turbine environment, related to flame holding, lean flame blow out, and thermoacoustic instabilities [10]. Regarding the latter, an experimental study suggests that an increased CO_2 content in the product gas can have a stabilizing effect [18]. Multiple studies suggest that already a small content of hydrogen in the order of 5% in volume may significantly improve flame stability [20,21].

This work is part of a project with the purpose of establishing commercially feasible power and heat production from LCV gas combustion in gas turbines. Toward this goal, a generic gas turbine burner is operated on a low LHV biosyngas with variable fuel composition, moisture content, and thermal power. The exhaust gas is analyzed in terms of emissions of CO and NO_x. The chemiluminescence of excited OH* radicals is imaged to assess the flame position and shape. Additionally, temperatures are measured at the combustor wall and at several positions inside the exhaust pipe.

Experimental Setup

Measurement Methods. The emissions in the exhaust flow were continuously measured. At the end of the exhaust tube, a sample was extracted from the exhaust gas and fed through a heated hosepipe into a cooled steam trap to remove humidity, before measuring the concentrations of unburned hydrocarbons, NO_x, CO, O_2, and CO_2 on a dry basis. Optical methods based on the pollutant's absorption behavior were used for the CO, CO_2, and NO_x measurements. The analyzer further distinguished between NO and NO_2. The measurement of O_2 utilized the molecule's paramagnetic characteristic.

To image the OH^* chemiluminescence of the flame, an image-intensified CCD camera was equipped with a UV-transmissive lens (f/3.8, $f = 78$ mm). Further, an interference filter, which was transmissive between 295 and 340 nm, was employed. A total of 480 frames at a frequency of 5 frames per second were taken for each operating point and the averaged OH^* distribution was calculated.

Temperatures were measured at various locations in the test rig using K-type thermocouples. This included the temperatures of the air and steam supply after heating, the burner inlet temperature, and the temperatures at the beginning and end of the exhaust tube. No correction for the heat radiation of the thermocouple was applied in the presented temperature measurements.

Atmospheric Combustion Test Rig. A sketch of the test rig is presented in Fig. 1. The burner used in this work has previously been developed at TU Berlin for the purpose of flashback-safe operation with high-reactivity fuels such as hydrogen under dry and wet conditions [22–30].

The LCV fuel gas is injected into the air and steam flow far upstream of the burner to ensure good mixing. The gas mixture is then fed to a radial swirler. The swirl number of the burner was adjusted by controlling the mass flux through the radial swirler vanes. For the results presented here, a constant geometrical swirl

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number of $S = 0.9$ was used (defined according to Ref. [31]). This swirl number ensured a strong recirculation zone and low velocities in the flame zone, which was favorable in terms of flame stability. Downstream of the radial swirl generator, a tube guided the premixed gases into a cylindrical quartz glass combustion chamber of 200 mm diameter and a length of 300 mm.

The flow field generated by the burner was thoroughly investigated in previous studies. A sudden area expansion at the combustion chamber inlet was used to induce a vortex breakdown and stabilize the flame in the shear layer between the central and corner recirculation zones [22]. Successful combustion of premixed hydrogen-air and natural gas–air with high contents of steam was also confirmed and led to extremely low emissions [23]. The burner concept has been also tested at high pressure and the positive effect of steam injection on low NO_x emissions was confirmed, especially at higher pressure levels, without negative effects on CO [29]. The combustor flow field was also explicitly investigated in water-tunnel studies by means of particle image velocimetry. A recent numerical simulation of the reacting flow with lean-premixed hydrogen confirmed the experimental results [30]. The thermoacoustic instability characteristics were also investigated and burner and flame transfer functions are available [32].

For the experiments in this work, methane was substituted with natural gas from the public gas grid. The LCV gas composition is varied by changing the methane (natural gas), steam, nitrogen, and $CO/CO_2/H_2$ concentrations. The relative hydrogen, carbon monoxide, and carbon dioxide content was fixed, since a bottled mixture of $CO/CO_2/H_2$ is used. The bottled mixture is supplied at a measured concentration of 35.6/18.9/45.5 vol % of $CO/H_2/CO_2$ (measurement uncertainty of ±2% relative). An overview of the gas supply hardware is shown in Fig. 2. Air, N_2, and steam are preheated and mixed upstream of the fuel injector port. The fuel is premixed from $CO/CO_2/H_2$ and natural gas and injected at room temperature. A second natural gas supply is used for ignition, but switched off for the remainder of the experiments.

Operating Conditions. The steam ratio Ω is defined as the ratio of steam to air mass flow

$$\Omega = \frac{\dot{m}_{\text{steam}}}{\dot{m}_{\text{air}}} \tag{1}$$

The equivalence ratio ϕ is defined as the fuel to air mass flow ratio normalized by the respective stoichiometric condition

$$\phi = \frac{\dot{m}_{\text{fuel}}/\dot{m}_{\text{air}}}{(\dot{m}_{\text{fuel}}/\dot{m}_{\text{air}})|_{\text{st}}} \tag{2}$$

The design point (reference case for a parameter study) is defined in Table 1. The product gas from a gasifier is experimentally

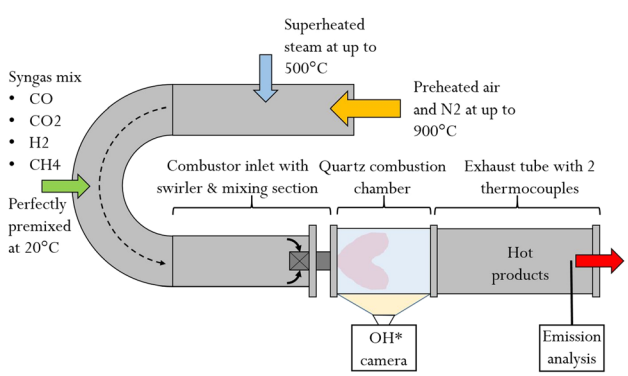

Fig. 1 Test rig sketch

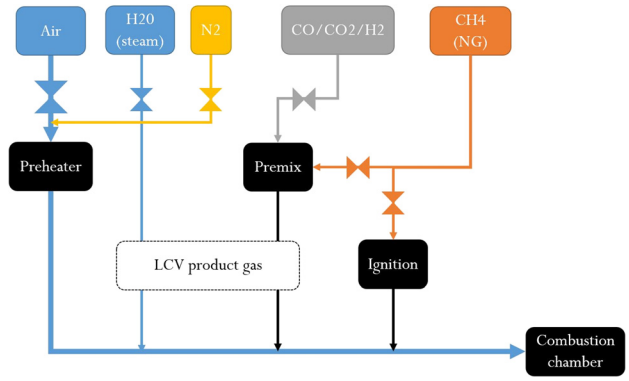

Fig. 2 Gas supply configuration

modeled (further referred to as biosyngas) and consists of the components steam, N_2, CH_4, and $CO/H_2/CO_2$. Table 1 presents the mass fraction w, volume fraction x at ambient condition, as well as the respective LHV. At the design point, the burner is operated at stoichiometric conditions, $\phi = 1$.

Starting with the design case, a sensitivity analysis was conducted by varying the flame thermal power P_{th}, steam ratio Ω, and fuel composition. Table 2 lists the range of values for the different parameters.

Results

The Results section has four parts. After discussing the flame shape and the blowout limits in the first two parts, the following two parts focus on the NO_x and CO emissions.

Flame Shape. The impact of steam admixture on the flame is visualized in Fig. 3, where a natural gas flame (top row) is compared to two biosyngas compositions for varying contents of steam. The biosyngas flame presented in the middle row has a higher methane content (15 vol % dry) compared to the design case (as specified in Table 1). The latter is displayed in the bottom row. In all cases the burner operated at stoichiometric conditions ($\phi = 1$) and at a constant thermal power of 60 kW. As a result, decreasing the methane content of the biosyngas is accompanied with a rising hydrogen content to keep the flame power constant. The figure shows the time-averaged OH^* chemiluminescence inside the combustion chamber normalized by the maximum value for each case.

The flame shape is of V-type and similar for all studied fuels at the lowest steam content of $\Omega = 0.3$ (left column in Fig. 3). The flame is compact and located close to the burner inlet. This is typical for a swirl-stabilized flame that anchors in the shear layer between the central and outer recirculation zones, where the local flow velocity and the turbulent burning velocity are equal [26]. The results show that even for the biosyngas composition at design conditions, the burning velocity is sufficiently high to

Table 1 Fuel composition at the design point

Component	w (%)	x (%)	LHV (MJ/kg)
Steam	46.2	57.0	—
N_2	24.8	19.7	—
Natural gas	3.3	4.5	49.85
$CO/H_2/CO_2$	25.7	18.8	4.85
–CO	8.5	6.7	10.16
–H_2	0.3	3.6	120.09
–CO_2	16.9	8.5	—
Total	100	100	2.87

Gray components: premixed $CO/H_2/CO_2$ broken down into individual species.

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Table 2 Parameters used in the experiments

Parameter	Value
Pressure	Atmospheric
Burner inlet temperature (°C)	470
Thermal power, P_{th} (kW)	45–85.7
Steam ratio, Ω	0.2–0.75
Methane content in biosyngas (vol % dry)	5–15
Hydrogen content in biosyngas (vol % dry)	7.5–9.3

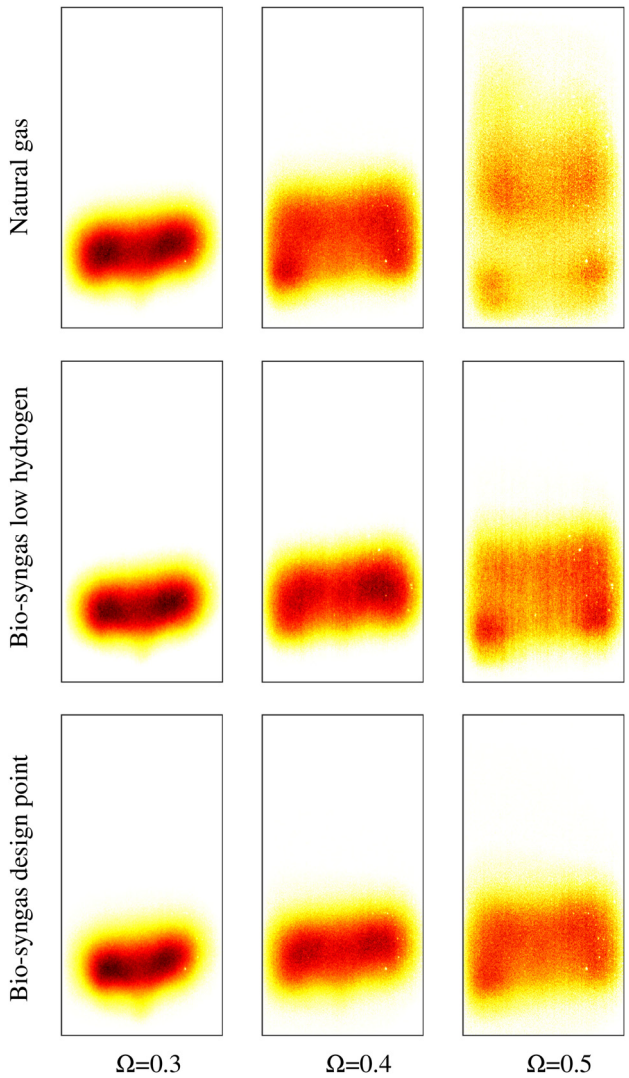

Fig. 3 Normalized time-averaged OH*-chemiluminescence for three different fuels at stoichiometric conditions and a constant thermal power $P_{th} = 60$ kW. Top to bottom: natural gas, biosyngas with low hydrogen content (7.5% hydrogen in dry biosyngas), and design point composition (8.4% dry). Each frame shows the entire combustion chamber with the burner inlet at the bottom.

allow the flame to stabilize in the shear layers. It is also worth noting however that for all three fuels the flame is asymmetric, with the flame being shifted slightly downstream on the right-hand side.

When the steam content is increased, the flame anchors slightly further downstream and appears less compact. This is expected, since an increase in steam admixture leads to a decrease of the reaction rate and flame speed [33]. This effect is stronger with increased natural gas content, as the simultaneously reduced hydrogen content further reduces the reactivity of the fuel. As a consequence, the reaction zone is lengthening axially, as well as radially. At a steam content of $\Omega = 0.5$, the pure natural gas flame reaches an unstable condition, where the flame pulsates between two quasi-stable points. This leads to two axial flame positions in time-averaged OH^* chemiluminescence, with one being close to the burner inlet and one being approximately half way down the length of the combustion chamber. With decreasing natural gas content, and increasing hydrogen content, respectively, this behavior is less pronounced and the flame appears more stable and anchors closer to the burner inlet. At design point conditions, the flame shape is only slightly broader than for the $\Omega = 0.4$ case. This indicates an increased flame speed of the biosyngas mixture containing hydrogen compared to the pure natural gas flame. Previous studies revealed that even if only a small amount of hydrogen is added to the fuel, flame speed and stability are increased [15,20,21].

Blowout Limit. The blowout limit has been assessed for different biosyngas compositions. Besides steam, the biosyngas composition is high in N_2 and CO_2, both increasing the overall dilution and contributing to flame cooling effects. The blowout limit of the biosyngas flame has been assessed by increasing the steam content at stoichiometric conditions up to the point of flame extinction, for several thermal power values and compositions. As shown in Fig. 3, the natural gas flame reaches an unstable point at $\Omega = 0.5$ and $P_{th} = 60$ kW. This condition was observed to precede flame extinction.

The tests revealed no significant influence of the thermal power on the blowout limit. Flame blowout occurred for the design fuel case at $\Omega = 0.65$ for a thermal power of $P_{th} = 45$ kW, at $\Omega = 0.68$ for $P_{th} = 60$ kW, and at $\Omega = 0.675$ for $P_{th} = 70$ kW, respectively. The slight decrease of the blowout limit in terms of Ω with decreasing thermal power is likely due to the increased relative influence of cooling by the combustion chamber walls.

Keeping the thermal power constant at $P_{th} = 60$ kW and decreasing the natural gas content while increasing the content of $H_2/CO/CO_2$ pushed the blowout limit to higher values of Ω (increased steam content). For the largest hydrogen content (9.3 vol % dry) flame blowout only occurred for $\Omega > 0.75$. This trend supports the assumption of increased flame speeds and reaction rates at elevated hydrogen content.

NO_x Emissions. The impact of the steam content on the dry NO_x-emission normalized to 15 vol % O_2 in the exhaust gas are shown in Fig. 4. The NO_x emissions of the biosyngas flame are compared to the emissions of a natural gas flame, for several thermal power values and fuel compositions of the former. All presented cases show very low, single-digit NO_x emissions. Furthermore, as expected, NO_x emissions decrease with increasing steam content.

NO_x emissions are sensible to the flame temperature [23]. For greater Ω and a constant air mass flow, the steam mass flow increases, which is expected to increase flame cooling. It can further be assumed that the additional dilution by CO_2 and N_2 in the biosyngas further amplifies this effect. In addition to the flame cooling effect, the presence of steam has an impact on the chemical reaction and plays a role in NO_x reduction as reported by Göckeler [34]. As a result, the NO_x emissions of all biosyngas flames are roughly independent of the flame power and the fuel composition, with slightly larger emissions for the biosyngas with increased natural gas content. As expected, the NO_x emissions for the natural gas flame are significantly higher compared to the biosyngas flames.

Figure 5 depicts the NO_x emissions for different biosyngas compositions and varying equivalence ratio. At each equivalence ratio, the flame temperature was held constant by adjusting the steam content to remove the temperature influence on the emissions. Again, the composition at the design point (in the following

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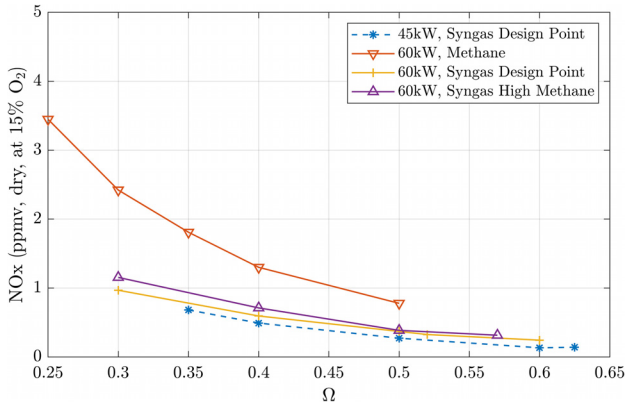

Fig. 4 NO_x emission over steam ratio Ω for several flames at $\phi = 1$

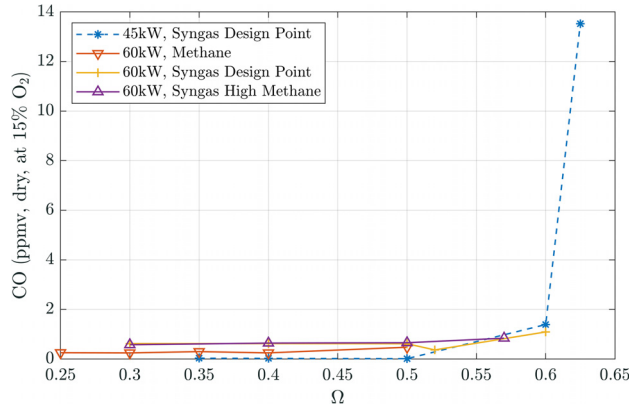

Fig. 6 CO emission over steam ratio Ω for several flames at $\phi = 1$

medium methane content) is compared to a composition containing a higher and lower methane content, and lower and higher hydrogen content, respectively. The flame power was $P_{th} = 60$ kW for all cases.

The NO_x emissions are roughly constant for all conditions measured here. A change in composition does not have a noticeable impact on the NO_x emission. Overall, a change in NO_x emission seems to be mainly influenced by a variation in flame temperature.

CO Emissions. Figure 6 presents the CO emissions for different contents of steam Ω and the same fuel compositions shown in Fig. 4. For most compositions, the normalized CO emissions are close to zero over a wide range of Ω.

For the biosyngas flames, CO emissions increase toward higher values of Ω. This is most pronounced for the 45 kW flame, where the flame became highly unstable. This is likely due to reduced flame speeds as a result of the increased steam content, which negatively affected flame stability. At these points, the CO emissions were increased and oscillated periodically. This effect was amplified with increasing steam content.

As for the NO_x emission, the CO emissions are compared for different compositions at equal flame temperatures in Fig. 7. The emissions for different fuel compositions follow a similar trend. At low values of ϕ lower CO emissions were detected, which increase when approaching the stoichiometric condition. It was observed that the flame became slightly unstable at high values of ϕ due to high values of Ω. Since for a constant flame power ϕ is adjusted via the air mass flow, a low equivalence ratio is synonymous with high air mass flows and hence low steam mass flows, and vice versa. The decreasing amount of oxygen surplus with rising ϕ and Ω might be an additional driving factor for elevated CO emissions.

Numerical Validation. Toward the goal of confirming the feasibility of the measured data, the equilibrium emissions of the biosyngas are simply computed by a reactor system comprising a zero-dimensional plug flow reactor (PFR) and a one-dimensional-flame reactor using the GRI-MECH 3.0 mechanism in Cantera [35].

The one-dimensional-flame reactor contains the flame front information such as the composition of the product gas and the adiabatic flame temperature and simulates the main flame reaction zone. This information is fed to the PFR, which mimics the heat loss during a long flame residence time in the main flame region. The PFR is composed of 2000 partially stirred reactors (PSRs) and models the postflame zone and the exhaust tube. The hot flue gas is brought to a new chemical equilibrium at temperatures present at the outlet of the exhaust tube. Here, temperatures of around 1000 K are measured in the experiment. The heat loss in the PFR is expressed as $Q = h_{tot}A(T - T_{env})$, where h_{tot} is the total heat transfer coefficient of the quartz glass combustion chamber and the exhaust tube and A is the surface area of each reactor. A list of the simulation parameters is presented in Table 3. Mass flow rates are taken from the experimental data.

As shown in Figs. 5 and 7, the simulation results show a reasonably well agreement with the experimental data except for a severe CO over-prediction at an equivalence ratio of $\phi = 1$. The simulation results imply an almost complete CO burnout until the end of the exhaust tube as shown by the measured low CO

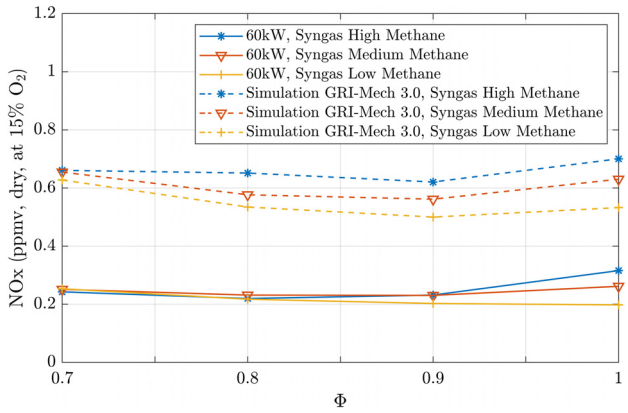

Fig. 5 NO_x emission and simulation data over equivalence ratio ϕ and corresponding steam ratio Ω at constant adiabatic flame temperature and flame power

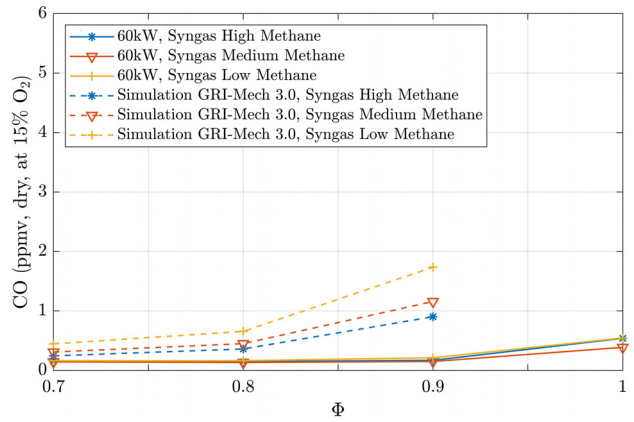

Fig. 7 CO emission and simulation data over equivalence ratio ϕ and corresponding steam ratio Ω at constant adiabatic flame temperature and flame power

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Table 3 Parameters used in the simulation

Parameter	Value
Length of postflame zone	0.2475
Length of exhaust tube (m)	0.865
Length of PFR (m)	1.1125
Surface area of single PSR (m^2)	1.9223×10^{-4}
Volume of single PSR (m^3)	5.046×10^{-6}
Flame residence time in PFR (ms)	61–67
Total heat transfer coefficient (W/m^2)	100

emission. For cases with an equivalence ratio close to $\phi = 1$, the associated high steam content greatly changes the reacting flow behavior and flame shape as it is shown in Fig. 3 and lead to the severe CO over-prediction. This can also explain the slight over-prediction of CO at other equivalence ratios. For lean combustion the over-prediction is less significant, since sharp V-flames feature lower residence time. At equivalence ratios close to unity, CO emissions from simulations are known to show a sensitive behavior to changes in ϕ. Reducing ϕ from 1 to 0.99 can lead to a reduction in CO emission by 50%.

Conclusions

This work presents the successful realization of stable combustion of artificially mixed low calorific value fuel gases employing a proprietary swirl-stabilized gas turbine burner.

A stable biosyngas flame was sustained over a wide range of steam contents Ω. Increasing the hydrogen content of the mixture improved flame stability, despite increasing the fraction of N_2 and CO_2 in the fuel at the same time. This is likely due to an increased flame velocity with increasing hydrogen content. However, its impact on the flame stabilization as reported in the literature still lacks the confirmation by simulations for this particular biosyngas composition.

The biosyngas mixture exhibited a LHV of 2.87 MJ/kg (2.7 MJ/m_N^3) at its target point. However, even lower LHV mixtures were shown to be combustible despite further increasing the steam content.

NO_x and CO emissions were analyzed for several thermal powers and fuel compositions. The biosyngas flames were emitting less NO_x compared to the methane flame at equal steam ratio Ω. This is likely due to the presence of N_2 and CO_2 and hence increased flame cooling. For all configurations emission values were in the single-digit NO_x range. A change in thermal power and fuel composition did not lead to a noticeable effect on the NO_x emissions, whereas the flame temperature was identified to exert a large leverage.

CO emissions are close to zero over a wide range of Ω for most operating points. Destabilization of the flame through increased steam or air dilution led to increased CO concentration in the exhaust.

Flame blowout occurred at higher Ω with increasing hydrogen content in the fuel. The flame power had only a slight impact on the blowout limit, potentially due to wall cooling effects at lower flame powers.

As a next step, pollutants commonly present in biogas as ammonia and tars will be added to the biosyngas to assess their impact on emission formation and combustion behavior.

Further, the development of a simulation tool toward predicting unstable points and hence operational points exerting increased CO emission is one of the key works in the next step.

Acknowledgment

This work was carried out as part of the project "BTC Pilot Plant, Phase 1" (No. 45487-1). The financial support of Phoenix Bio-Power AB and the Swedish Energy Agency is gratefully acknowledged.

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